SG1: Visualizing the Pill Puzzle

Test Plan Document

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Team Members:

- **Daniel McKinnis**
- **AJ Soma Ravichandran**
- **Matthew Yeager**
- **Jacob Young**

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Table of Contents

1. Introduction …….. 3

2. Testing Approach …… 3

3. Test Cases …… 4-10

4. Test Execution Log ……. 11

Introduction

This document describes the testing performed for the SG1 program. The purpose of this document is to record the test cases used to verify that the program behaves as expected when users enter different inputs.

The tests focus on verifying:

- Correct handling of valid inputs
- Correct handling of invalid inputs
- Correct program output printed to the screen

Testing Approach

The testing approach involves manually running the program and entering specific inputs when prompted. Each test case records:

- The exact input provided by the user
- The expected output
- The actual output produced
- The date and time of execution
- The tester
- The test result (Pass / Fail)

Test Cases

Test Case 1

Description: Test with N = 3, R = 10 and D = 7

Exact User Input: 3 10 7

Expected Output: Expected to have zero pills on day 7, as 3 pills = 6 half pills and that's only 6 days. Expect question 2 to output min, max, average, and most likely day where the last whole pill was taken correctly.

Execution	**Date**	**Time**	**Tester**	**Actual Output**	**Result**
N = 3, R = 10, D = 7	03/05/2026	09:38 am	Matthew Yeager	Continues to prompt for the correct D value. The number of days that there are pills = 2*N (each day take half a pill)	This worked as expected. It showcases the bounds checking for inputs. Technically this test passed, as it correctly prompts the user for a correct input instead of crashing.

```
SG1: Pill Puzzle Simulator
This program simulates the 'Pill Puzzle' R times for a bottle starting with N whole pills.
It shows progress while running and collects statistics for later display.

Enter the number of whole pills (N, 1-1000): 3
N accepted: 3 (the bottle starts with 3 whole pills)

Enter the number of simulations (R, 1-10000): 10
N accepted: 10 (the simulation will run 10 times)

Enter the day to analyze (D, 1-6): 7
  Error: Value out of range. Please enter a number between 1 and 6 (inclusive).
Enter the day to analyze (D, 1-6): 7
  Error: Value out of range. Please enter a number between 1 and 6 (inclusive).
Enter the day to analyze (D, 1-6): 7
  Error: Value out of range. Please enter a number between 1 and 6 (inclusive).
Enter the day to analyze (D, 1-6):
```

Test Case 2

Description: Test with N = 5, R = 10, and D = 3

Exact User Input: 5 10 3

Expected Output: Expected to work correctly. Day 3 is appropriate for this scenario and each number is valid

Execution	**Date**	**Time**	**Tester**	**Actual Output**	**Result**
N = 5, R = 10, D = 3	03/05/2026	09:45 am	Matthew Yeager	Q1: Expected whole pills on day D: 2.6 Expected half pills on day D: 1.8 Q2: Most likely day: 7 Q3: Probability of drawing a half pill on day D: .60	Worked as expected. Everything outputted correctly. Passed.

```
Enter the number of whole pills (N): 5
Enter the number of simulations (R): 10
Enter the day to analyze (D): 3
Progress: |                                        | 100.0%

--- Question 1: Expected pill counts on day D ---
  Expected whole pills: 2.6000
  Expected half  pills: 1.8000

--- Question 2: Analysis of Last Whole Pill Day ---
  Minimum day: 6
  Maximum day: 8
  Average day: 7.3000
  Most likely day: 7

Displaying histogram...

--- Question 3: Half-draw probability on day D ---
  Estimated probability of drawing a half pill on day 3: 0.600000
```

Test Case 3

Description: Test with alphabetical character for each input

Exact User Input: N = A, R = A, D = A

Expected Output: Expects to prompt user for number inside range for each value

Execution	Date	Time	Tester	Actual Output	Result
N = A, R = A, D = A	03/06/2026	09:38 am	Matthew Yeager	Prompts user for whole number between given range.	Passed. Continues to prompt for the correct character instead of running and crashing.

```
Enter the number of whole pills (N, 1-1000): A
  Error: Invalid input. Please enter a whole number (no letters, decimals, or symbols).
Enter the number of whole pills (N, 1-1000): 5
N accepted: 5 (the bottle starts with 5 whole pills)

Enter the number of simulations (R, 1-10000): A
  Error: Invalid input. Please enter a whole number (no letters, decimals, or symbols).
Enter the number of simulations (R, 1-10000): 5
N accepted: 5 (the simulation will run 5 times)

Enter the day to analyze (D, 1-10): A
  Error: Invalid input. Please enter a whole number (no letters, decimals, or symbols).
Enter the day to analyze (D, 1-10): 5
N accepted: 5 (day 5 will be checked)
```

Test Case 4

Description: Test the program with out-of-range values for N, R, and D.

Exact User Input: N = 20000, R = 90000, D = 90

Expected Output:
The program should display an error message indicating that the entered values are outside the allowed range for the number of whole pills (N), the number of simulations (R), and the day (D). The program should then repeatedly prompt the user to enter a valid number within the specified range until a correct value is provided.

Execution	**Date**	**Time**	**Tester**	**Actual Output**	**Result**
N = 20000, R = 90000, D = 90	03/08/2026	6:48 pm	AJ Soma Ravichandran	The program displayed an input error message and re-prompted the user to enter a valid number for N, R and D.	Passed

```
Enter the number of whole pills (must be between 1 and 1000): 20000
Error: Value out of range. Please enter a number between 1 and 1000 (inclusive).

Enter the number of whole pills (must be between 1 and 1000): 34
Input accepted: 34 whole pills will be placed in the bottle.

Enter the number of simulations to run (must be between 1 and 10000): 90000
Error: Value out of range. Please enter a number between 1 and 10000 (inclusive).

Enter the number of simulations to run (must be between 1 and 10000): 800
Input accepted: The simulation will run 800 times.

Enter the day to analyze (1–68): 79
Error: Value out of range. Please enter a number between 1 and 68 (inclusive).

Enter the day to analyze (1–68): 90
Error: Value out of range. Please enter a number between 1 and 68 (inclusive).

Enter the day to analyze (1–68): 34
Input accepted: Statistics will be calculated for day 34.
```

Test Case 5

Description: Test the program with a decimal value for N, an empty value for R, and a negative number for D.

Exact User Input: N = 0.9, R = (empty value) and D = -89

Expected Output:
The program should display an invalid input error message indicating that letters, decimals, negative numbers, or symbols are not allowed when entering N, and it should re-prompt the user to enter a valid integer.

For R, the program should display a “No value entered” error message and re-prompt the user for input.

For D, the program should display the same invalid input error message as for N, since the entered value is a negative number, and then re-prompt the user for a valid integer within the allowed range.

Execution	**Date**	**Time**	**Tester**	**Actual Output**	**Result**
N = 0.9, R = (empty value) and D = -89	03/08/2026	6:59 pm	AJ Soma Ravichandran	The program displayed the error message **“no letters, decimals, negative**	Passed the test, re-prompted as expected

				numbers, or symbols allowed” for **N** and **D**, and **“No value entered”** for **R**, then re-prompted the user for valid input.	

```
Enter the number of whole pills (must be between 1 and 1000): 0.9
Error: Invalid input. Please enter a whole number (no letters, decimals, negative numbers or symbols).

Enter the number of whole pills (must be between 1 and 1000): 80
Input accepted: 80 whole pills will be placed in the bottle.

Enter the number of simulations to run (must be between 1 and 10000):
Error: No value entered. Please enter a whole number.

Enter the number of simulations to run (must be between 1 and 10000): 90
Input accepted: The simulation will run 90 times.

Enter the day to analyze (1–160): -89
Error: Invalid input. Please enter a whole number (no letters, decimals, negative numbers or symbols).

Enter the day to analyze (1–160):
```

Test Case 6

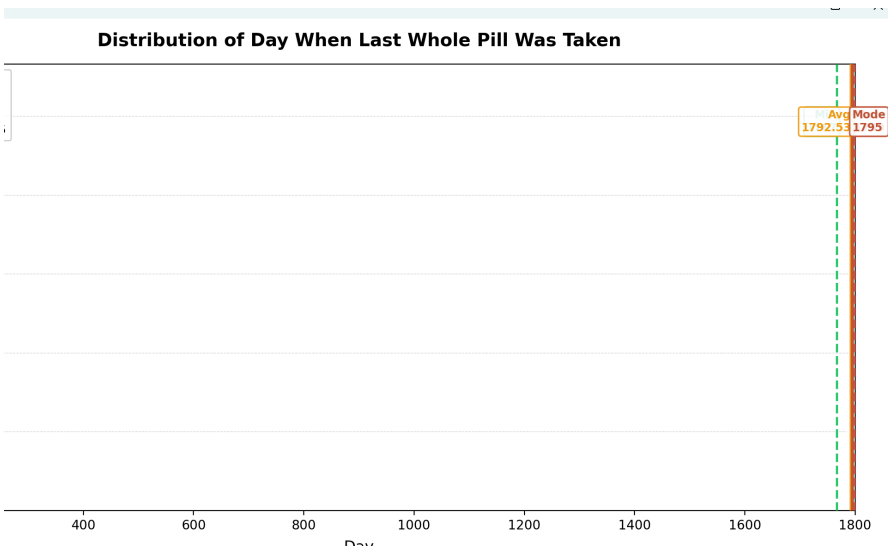
Description: Test with higher valid input values for N, R and D, test for histogram visualization, results for 3 questions, sequences for experts section, “enter” prompt for end the

Exact User Input: N=900, R=900, N = 80

Expected Output: Displays a progress bar during calculations, then shows results for all three questions, a histogram visualization, and an ENTER prompt to end the program.

Execution	Date	Time	Tester	Actual Output	Result
N=900, R=900, N = 80	03/08/2026	7:08 PM	AJ Soma Ravichandran	Displayed a progress bar during simulations, printed results for all three questions, showed the visualization, and prompted the user to press ENTER to end the program.	Passed

```
Enter the number of whole pills (must be between 1 and 1000): 900
Input accepted: 900 whole pills will be placed in the bottle.

Enter the number of simulations to run (must be between 1 and 10000): 900
Input accepted: The simulation will run 900 times.

Enter the day to analyze (1–1800): 80
Input accepted: Statistics will be calculated for day 80.

Progress: |                                        | 100.0%

--- Question 1: Expected pill counts on day D ---
  Expected whole pills: 823.4056
  Expected half  pills: 73.1889

--- Question 2: Analysis of Last Whole Pill Day ---
  Minimum day: 1750
  Maximum day: 1799
  Average day: 1792.5022
  Most likely day: 1798

Displaying histogram...
```




```
--- Question 3: Half-draw probability on day D ---
  Estimated probability of drawing a half pill on day 80: 0.092222
==================================================
      Running Sequence Probability: N=4, R=100000
==================================================
Sequence      | Occurrences  | Probability
--------------------------------------------------
WWWHWHHH      | 9386         |   9.39%
WWWWHHHH      | 9372         |   9.37%
WWWHHHWH      | 9368         |   9.37%
WWWHHWHH      | 9339         |   9.34%
WWHWHHWH      | 8408         |   8.41%
WWHWHWHH      | 8333         |   8.33%
WWHWWHHH      | 8257         |   8.26%
WWHHWWHH      | 6297         |   6.30%
WWHHWHWH      | 6281         |   6.28%
WHWWHHWH      | 5663         |   5.66%
WHWWHWHH      | 5589         |   5.59%
WHWWWHHH      | 5451         |   5.45%
WHWHWHWH      | 4152         |   4.15%
WHWHWWHH      | 4104         |   4.10%

All simulations completed.
Press ENTER to end the program.
```

Test Execution Summary

Total Tests	Passed	Failed
6	6	0